

From Parity Certificates to a Community Network: Exact Image Decoding and Sharper Child Precision

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AI-assisted research and formalization

8 October 2026

Abstract

We combine Gaussian parity certificates with the mixed-center scalar producer, copied-center schedule, phase endpoint, and semantic precision interfaces in the community’s strongest available integer-multiplication candidate, PR46. The resulting contribution is a working exact reference kernel and checked interface: subtract dirty old/new values in a common frame, decode the fresh integer image, and collect side corrections before division. This eliminates six extra fractional bits in the compared scalar reference schedules. A complete producer–decoder round trip checks image membership. Separately, the actual 29-class complex child profile admits a 49.8644% smaller completed-child precision charge. Lean proves the arithmetic contracts, canonical denominator normalization, recursive tensor execution for every dimension, inverse restoration, and profile bounds; an independent replay checks all 10,732,176 scalar coefficients. These results do not establish a new multiplication exponent or a faster physical tape implementation.

1 Community synthesis and the contribution boundary

The source-pinned inventory covers all 46 pull requests available at the start of this one-hour investigation, plus Swapnil Jain’s separate Round6. The latest candidate, PR46, combines RaD/hipotures’s changed DAGs, fixed local bases, reversed corners, copied retained centers, Rohan Arun’s weighted carrier matching, and Chafik Boukhalfa’s exact rational recovery of ten exceptional pairs. Its conditional witness is $\kappa = 82012589/(2 \cdot 10^{12}) = 4.10062945 \cdot 10^{-5}$. These techniques are already composed; their savings cannot simply be added. We pin PR46 at 71b6c960c89295e952522dd44111df9cfc51ae90.

The present increment supplies an executable arithmetic refinement for that construction. Its starting point is the earlier Archivara parity-aware multiplication exploration, subsequently developed personally by the author with multiple AI tools, most recently GPT-6.1 according to the author’s account. The new work, including code and Lean, involved substantial OpenAI Codex assistance. We use the earlier line’s motivation to preserve arithmetic congruences, and independently prove every imported arithmetic claim. We do not import its broad circuit-separation or benchmark assertions.

2 Exact decoding with dirty auxiliaries

Let $\mathcal{T}$ be the triples of $\{0, \dots, 27\}$, so $|\mathcal{T}| = 3276$, and let $\mathcal{D} = \{0, \dots, 18\}$. For arbitrary Gaussian integer input values X_S , set

$$T = \sum_{S \in \mathcal{T}} X_S, \quad G_j = \sum_{S \ni j} X_S, \quad D_j = T - G_j \ (j \in \mathcal{D}), \quad B_j = 2G_j \ (j \notin \mathcal{D}).$$

Each source occurs in exactly three G_j , giving

$$N = \sum_{j \notin \mathcal{D}} B_j - 2 \sum_{j \in \mathcal{D}} D_j = 2(3 - 19)T = -32T. \tag{1}$$

This is the mixed-center identity already used by the community. Our refinement attaches exact-division contracts to its producer image, and implements them in a scalar reference evaluator.

Theorem 1 (Dirty-safe image decoding). *Let P be the complete scalar producer, including its side ports. For arbitrary old auxiliary values a , set $a' = a + P(X)$. Then $\Delta = a' - a = P(X)$. On these differences, every $\Delta B_j/2$ and $\Delta N/(-32)$ is an exact Gaussian integer division. The recovered values are G_j and T . This assertion requires old/new values expressed in one common scalar frame across all producer ports.*

Proof. Subtracting cancels arbitrary a . Equation (1) and $B_j = 2G_j$ hold for each source column and hence for their sum, separately in the real and imaginary coordinates. For $j \in \mathcal{D}$, recover $G_j = T - \Delta D_j$. The Lean source proves the column identity, arbitrary finite superposition, exact division even for negative integers, and the dirty difference identities. No parity promise is imposed on a . $\square$

For target S , let A_S be the sum of inputs disjoint from S , and E_S the sum with intersection size two. The latter is assembled from the three actual pair-star ports. The community scalar scatter is

$$Q(PX)_S = \frac{\sum_{j \in S} G_j - T + A_S - E_S}{2}. \quad (2)$$

Our implementation collects the numerator before dividing.

Theorem 2 (Integral collected correction). *For every input $X \in \mathbb{Z}[i]^{3276}$, the numerator in (2) is $2X_S$. Thus the collected reference output is integral, with no added fractional bits. Before the side correction, one extra binary fractional bit can be necessary.*

Proof. For source R write $j = |R \cap S|$. Twice its coefficient in (2) is

$$j - 1 + [j = 0] - [j = 2] = 2[j = 3], \quad j \in \{0, 1, 2, 3\}.$$

Two triples intersect in three points precisely when they are equal. Lean proves the four cases and their arbitrary integer superposition. A disjoint unit source contributes $-1/2$ to the center-only scatter, showing why collection with the side correction matters. $\square$

A complete image check and failure controls

Successful divisibility tests alone do not characterize $\text{im } P$. Let Q be the complete decoder in (2), for which $QP = I$. Then

$$\Delta \in P(\mathbb{Z}[i]^{3276}) \iff Q\Delta \in \mathbb{Z}[i]^{3276} \text{ and } P(Q\Delta) = \Delta. \quad (3)$$

Here the displayed decoder is rational-valued off the producer image; the checked adapter requires all its divisions to be integer-exact and performs this round trip. A faster image-conditioned path can omit it only when the producer witness is already available. The generic equivalence, projection idempotence, and checked acceptance interface are Lean-proved. The finite producer-decoder link is independently verified for this actual network.

A single arbitrary B center equal to one yields $N = 1$ and an illegal division by -32 ; generic scatter outputs can require denominator 64. Conversely, a disjoint side port equal to two with all other ports zero passes local exact divisions but fails (3). A frame mismatch is also unsafe: rotating an unchanged old center from 1 to i creates apparent difference $i - 1$, which is not a valid fresh update. These are explicit rejection tests.

3 Canonical Gaussian arithmetic and the paid endpoint

Write $\varpi = 1 + i$, so $\varpi^2 = 2i$, and represent an exact value by (z, e) meaning z/ϖ^e , with $z \in \mathbb{Z}[i]$ and $e \geq 0$. The residue $(\Re z + \Im z) \bmod 2$ vanishes precisely when division by ϖ is exact. Repeated checked cancellation yields the unique canonical representative: either $e = 0$, or the numerator has nonzero residue. Zero canonically has tag zero. Lean proves termination via the decreasing tag, soundness, uniqueness, idempotence, and preservation of semantic equality.

For heterogeneous tags, first multiply numerators by the required powers of ϖ to reach their maximum tag. The exact phase kernel

$$C = \frac{iI + X}{\varpi}, \quad C^{-1} = \frac{I + iX}{\varpi}$$

always retains a sufficient denominator and then canonicalizes. Unlike a parity-guarded integer-only kernel, it is valid on every Gaussian dyadic value. Lean proves $C^2 = X$ and inverse restoration with arbitrary incoming tags, with literal representation restoration after canonicalization. Halving is $(z, e) \mapsto (iz, e + 2)$; phases preserve the tag. Gaussian denominator reduction is established prior work, notably Amy–Glaudell–Ross, section 5.4; we claim the specialized implementation and its checked interface, not a new number-theory principle.

The tensor arithmetic is now executed in Lean

The new `GaussianTensorExecution.lean` represents a dimension- D cube by a Gaussian integer at $D = 0$ and a pair of $(D - 1)$ -cubes otherwise. It recursively executes the two children and the new raw C axis; its inverse reverses the axis and then the children. For every cube z , the kernel checks the actual program identity

$$\text{inverse}_D(\text{forward}_D(z)) = \varpi^{2D} z$$

coordinatewise. Forward/inverse tags charge their actual $2D$ denominator units, so every returned value equals its input; canonicalization restores the literal representation. Terminal binary-grid conversion and the executed all-one input impulse are also proved, giving the sharp $p + \lceil D/2 \rceil$ allowance on incoming binary precision p . This closes the earlier paper’s written-only link for this recursive tensor arithmetic. It does not formalize the community’s framed physical schedule.

The community’s two-stage complex endpoint writes $F = C_I$, $T = C_U$, and $E = FT^{-1}$, with U the norm-one tensor-triple line of integer weight nine. Its physical data outputs are

$$A = -Fy, \quad B = FT^{-2}x + Ey.$$

The paid copied correction $T^{-1}A = -Ey$ leaves $FT^{-2}x$; the input/output phase wrappers satisfy $i^9 Z_U FT^{-2} Z_U = F$. Our new integer-based Gaussian interpreter checks these identities on heterogeneous dirty dyadics at dimensions 9, 10, and 11. This is a finite phase audit, not execution at the physical dimension 784. In the full physical invocation, dirty auxiliary endpoints are transformed by the required F ; scalar restoration after stripping frames must not be mistaken for unchanged physical dirty streams.

4 A sharper charge for the actual community child list

The earlier parity paper establishes the sharp completed-child binary grid $2^{-\lceil D/2 \rceil} \mathbb{Z}[i]$ for a D -axis C tensor. This remains valid after fourth-root phases and address permutations. In the component box norm, its amplification is at most $2^{\lceil D/2 \rceil}$. The complex-modulus row norm is $2^{D/2}$; these norms differ at odd depth. The normalized outer H_0 interface still requires D binary bits.

PR46’s 29-class complex child list has multiplicities n_r , with

$$s = \sum_r n_r r = 421548223824, \quad o = \sum_{r \text{ odd}} n_r = 1142904672.$$

When a child has width rf , its sharp completed binary allowance is $\lceil rf/2 \rceil$. Therefore its summed conservative charge is exactly

$$H(f) = \sum_r n_r \left\lceil \frac{rf}{2} \right\rceil = \frac{sf + o(f \bmod 2)}{2} \leq 211345564248f. \quad (4)$$

Relative to the former sf contribution, the uniform coefficient decreases by 210202659576, or 49.8644396%. Lean checks the pinned list, its rank mass and odd multiplicity, and the bound for every natural f . If a common ϖ tag is retained through a sequence and conversion occurs only once, one can instead convert the cumulative tag at the final boundary. The scalar halving charges must still be included.

Equation (4) improves a *component* of the semantic guard. The large scalar constant E , recursive unfinished-child bounds, physical layouts, and final normalization remain separately charged. It does not imply a 49.8644% reduction in total memory, execution time, or the published guard constant, and it does not alter the characteristic or κ .

5 Executable verification and next integration step

The delivered package contains a standalone Lean project, an exact Gaussian interpreter, the mixed-center reference kernel and complete image check, source snapshots with retained notices, and a community ledger. The independent producer replay checks all 81,885 nodes, including 78,437 active additions, all 13,132 root ports, and all 10,732,176 source/target coefficients with exact integer bitplanes. Six dirty-input/output trials compare all targets to independent rational old/new scatter and verify scalar auxiliary restoration. The compared reference schedules make 26,268 versus 3,286 Gaussian quotient calls, and require six versus zero extra fractional bits. These counts exclude the optional full image round-trip and all physical frame transport; they are not optimality or runtime claims.

The canonical Gaussian adapter additionally checks the actual producer and decoder on heterogeneous source and dirty denominator tags, at a promised common input grid. It recovers canonical inputs and restores canonical old auxiliaries literally. The delivered Lean project checks 168 theorem declarations in total, including 99 new ones beyond the earlier package.

The practical handoff is precise: use this as an exact reference and assertion boundary at a completed common-frame scalar shear, or implement the same rewrite after proving and charging old/new frame transport and temporary-stream scheduling. The current scalar implementation cannot be substituted into arbitrary physical gates without that argument. Its divisibility, canonicalization, endpoint controls and narrower child bound are available now for compiler development and independent review. The full upstream theorem, analytic recovery, fixed-tape execution, spectator parking, and eventual thresholds are outside this formalization. The upstream full `make verify` is not claimed to have been rerun.

References

- [1] Chafik Boukhalfa et al., community PR46, [weighted matching and exact data recovery](#), pinned commit above. Its source notices credit the retained contributors.
- [2] icekylinx, PR36, [copied retained centers](#); PR32 supplies the fixed-basis mixed-center producer.
- [3] RaD/hipotures, PR20; Zhihao Chen, PR23, [semantic precision and bulk interfaces](#).
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- [6] M. Amy, A. N. Glaudell, N. J. Ross, *Number-theoretic characterizations of some restricted Clifford+T circuits*, Quantum 4, 252 (2020), section 5.4, [doi:10.22331/q-2020-04-06-252](#).